

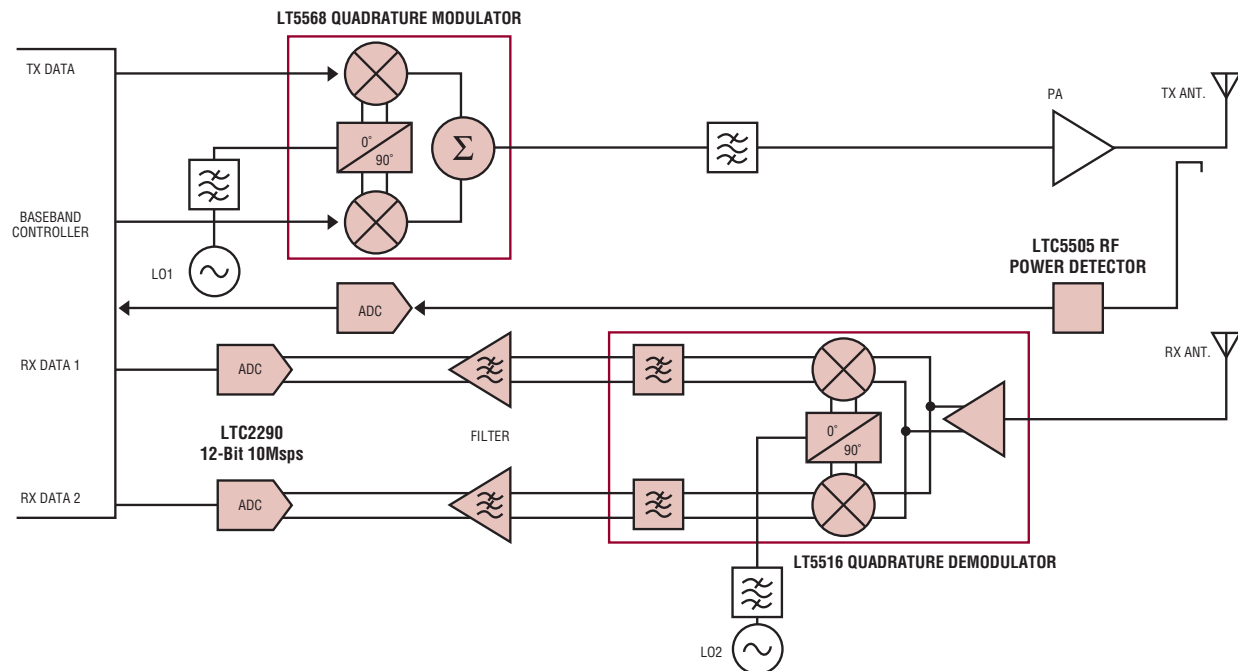
Linear Technology Chronicle

High Performance Analog Solutions from Linear Technology

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Focus...

Solutions for RFID



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Radio-frequency identification (RFID) is an automatic identification method that allows quick and reliable detection and recognition of objects ranging from cars to clothes to razor blades. Unlike older bar code or magnetic strip identification methods, RFID readers can identify tagged objects at a distance of up to several meters quickly and with no need for the object to be precisely aligned to the reader.

In most RFID systems, the tag is a simple passive device that receives its power through RF energy transmitted from the RFID reader. Using a backscattering technique, the tag communicates its unique identifying code back to the RFID reader. While individual systems vary, common frequency bands include 900MHz, 13.56MHz and some lower frequencies.

RFID system environments can be quite hostile. The frequency channels operate in a license-free ISM (industrial, scientific, and medical) band. RFID readers in this band

are prone to external interference from cordless telephones, wireless headsets, wireless data networks, and other co-located readers. Each reader's RF receiver front end must be designed to withstand high-interference signal levels without introducing distortion products that cause interrogation errors. The receiver noise needs to be low so that it has sufficient dynamic range to allow error-free detection of low-level responding tag signals.

Linear Technology designs and manufactures the RF and signal conditioning components required for high performance, high reliability RFID systems. Our quadrature demodulators and modulators, RF power detectors and filters offer the high performance, innovative design and unmatched reliability that design engineers expect from Linear Technology.

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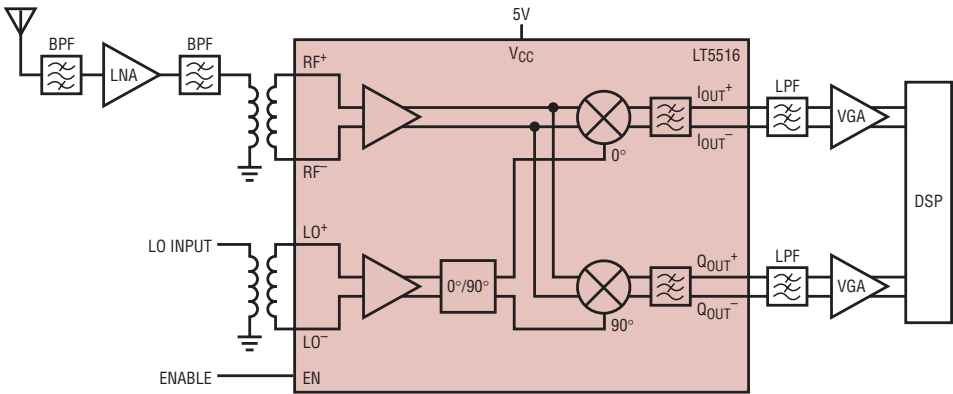
Quadrature Demodulators for RF Receivers

It is a significant engineering challenge to detect the small signal returned by the passive RFID tag in the presence of the large amplitude transmitted signal. Each reader's RF receiver front end must be designed to withstand high-interference signal levels without introducing significant distortion spurs. The receiver noise needs to be low enough that the system has sufficient dynamic range to allow error-free detection of low-level responding tag signals.

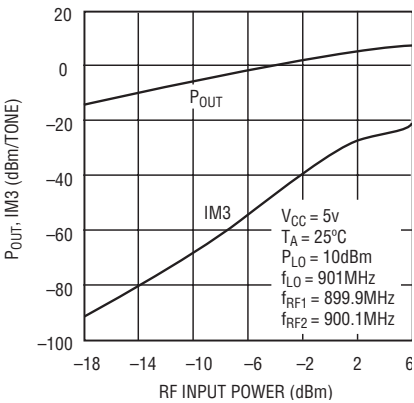
The LT®5516 is an 800MHz to 1.5GHz direct conversion quadrature demodulator optimized for high linearity receiver applications. The LT5516 incorporates balanced I and Q mixers, LO buffer amplifiers and a precision, high frequency quadrature generator. This direct conversion receiver can eliminate the need for intermediate frequency (IF) signal processing, as well as the corresponding requirements for image and IF filtering. Features of the part include:

- **Frequency Range:**
800MHz to 1.5GHz
- **High IIP3: 21.5dBm at 900MHz**
- **High IIP2: 52dBm**
- **Noise Figure: 12.8dB at 900MHz**
- **P1dB=9dBm**

High Signal-Level I/Q Demodulator for Wireless Infrastructure



I/Q Output Power IM3 vs RF Input



Quadrature Demodulators

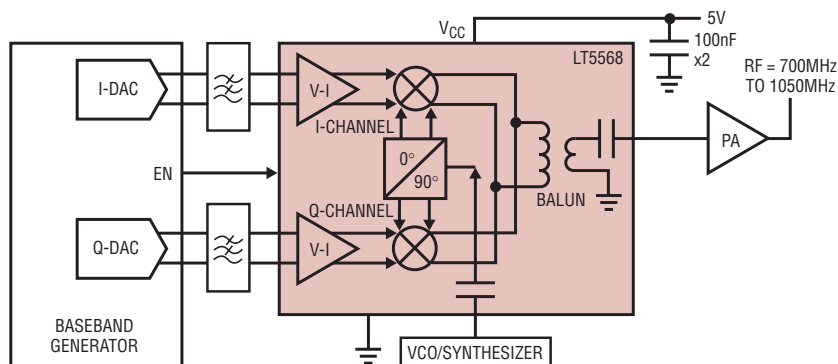
Part Number	Function	Op Frequency	BB Frequency	IIP3	NF	Conversion Gain	Ampl./Phase Match	V _{CC}	I _{CC}
LT5516	900MHz ISM	0.8 - 1.5GHz	DC - 260MHz	21.5dBm	12.8dB	4.3dB	0.2dB/1°	4.0V–5.25V	117mA
LT5515	2.4GHz ISM	1.5 - 2.5GHz	DC - 260MHz	20dBm	16.8dB	–0.7dB	0.3dB/1°	4.0V–5.25V	125mA
LT5517	13.56MHz ISM	10 - 900MHz	DC - 130MHz	23.5dBm	9dB	–4dB	0.03dB/0.7°	4.5V–5.25V	90mA

Quadrature Modulators for RF Transmitters

The LT5568 offers very high linearity and a low noise floor, providing exceptional dynamic range for the transmitted signals. The modulator accepts the baseband I and Q signals in quadrature from the digital-to-analog converters and modulates directly to the 900MHz transmit frequency. Features of the part include:

- **Frequency Range:**
700MHz to 1050MHz
- **High OIP3: +22.9dBm at 850MHz**
- **Low Output Noise: -160.3dBm/Hz**
- **LO Leakage = -43dBm**
- **Image Suppression = -46dBc**

700MHz to 1050MHz Direct Conversion Transmitter Application



Other LTC Quadrature Modulators

Part Number	Op Frequency	OIP3	Noise Floor	Baseband DC Input Voltage	Carrier Suppression	Image Rejection	V _{CC}	I _{CC}
LT5558	0.6-1.1GHz	22.4dBm	-158dBm/Hz	2.1V	-43.7dBm	-49dBc	4.5V - 5.25V	108mA
LT5518	1.5-2.4GHz	22.8dBm	-158.2dBm/Hz	2.06V	-49dBm	-40dBc	4.5V - 5.25V	128mA
LT5528	1.5-2.4GHz	21.8dBm	-159.3dBm/Hz	0.525V	-42dBm	-45dBc	4.5V - 5.25V	125mA

RF Power Detectors

High performance RF power detectors allow precise measurement and control of the transmitted signal power in the RFID system. The LT5534 is a 50MHz to 3GHz monolithic RF power detector capable of measuring RF signals over a 60dB dynamic range. The RF signal in a decibel scale is precisely converted into DC voltage on a linear scale. The LT5534 delivers

superior temperature stability (typical output variation within ± 1 dB over the full temperature range). The output responds in less than 40ns to a large RF input signal. Features of the part include:

- **RF Frequency Range:**
50MHz to 3GHz
- **Dynamic Range: 60dB**

- **Exceptional Accuracy over Temperature and Power Supply**
- **Fast Transient Response: 38ns**
- **Tiny 6-Lead SC70 Package**

RF Power Detectors for RFID

Part Number	Function	Op Frequency	Dynamic Range	Detect Range	V _{CC}	I _{CC}	Package
LT5534	Log-Linear	50MHz - 3GHz	60dB	-63 to +2dBm	2.7V - 5.25V	7mA	2mm x 2mm SC70
LTC5507	Low Frequency	100kHz - 1GHz	48dB	-34 to +14dBm	2.7V - 6V	550 μ A	6L ThinSOT™
LTC®5505	Low Cost	0.3 - 3GHz	44dB	-28 to +18dBm	2.7V - 6V	500 μ A	5L ThinSOT
LTC5508	Schottky Peak	0.3 - 7GHz	44dB	-32 to +12dBm	2.7V - 6V	580 μ A	2mm x 2mm SC70

Filters

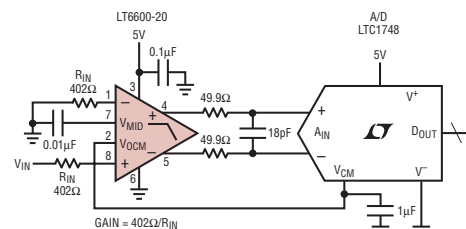
Before the received signal can be digitized by the baseband processor, it often needs additional external filtering. The LT6600 combines a fully differential amplifier with a 4th order lowpass with a fixed bandwidth of 2.5, 5, 10, 15 or 20MHz. The LT6600 also provides the necessary level shifting to set its output common mode voltage to accommodate the reference voltage requirements of A/Ds.

Using a proprietary internal architecture, the LT6600 integrates an antialiasing filter and a differential amplifier/driver without compromising distortion or low

noise performance. At unity gain, the measured in-band signal-to-noise ratio is an impressive 76dB. Features of the part include:

- **Programmable Differential Gain via Two External Resistors**
- **Adjustable Output Common Mode Voltage**
- **0.5dB Ripple 4th Order Lowpass Filter**
- **76dB SNR with 3V Supply and 2V_{P-P} Output**
- **SO-8 Package**

LT6600 Lowpass Filter



Selected Lowpass Filters

Part Number	Order	Cutoff Frequency	SNR	HD2	HD3	Ripple	Package	V _{CC}	I _{CC}
LT1568	4th	Prog. to 10MHz	92dB	-77dBc	-92dBc	N/A	2x2 SC70	2.7V - 11V	26mA
LT6600-2.5	4th	2.5MHz	86dB	-95dBc	-88dBc	0.5dB	6L SOT-23	2.7V - ±5V	30mA
LT6600-5	4th	5MHz	82dB	-93dBc	-96dBc	0.5dB	5L SOT-23	2.7V - ±5V	31mA
LT6600-10	4th	10MHz	82dB	-88dBc	-97dBc	0.5dB	5L SOT-23	2.7V - ±5V	39mA
LT6600-15	4th	15MHz	76dB	-86dBc	-90dBc	0.5dB	5L SOT-23	2.7V - ±5V	46mA
LT6600-20	4th	20MHz	76dB	-83dBc	-88dBc	0.5dB	2x2 SC70	2.7V - ±5V	46mA

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